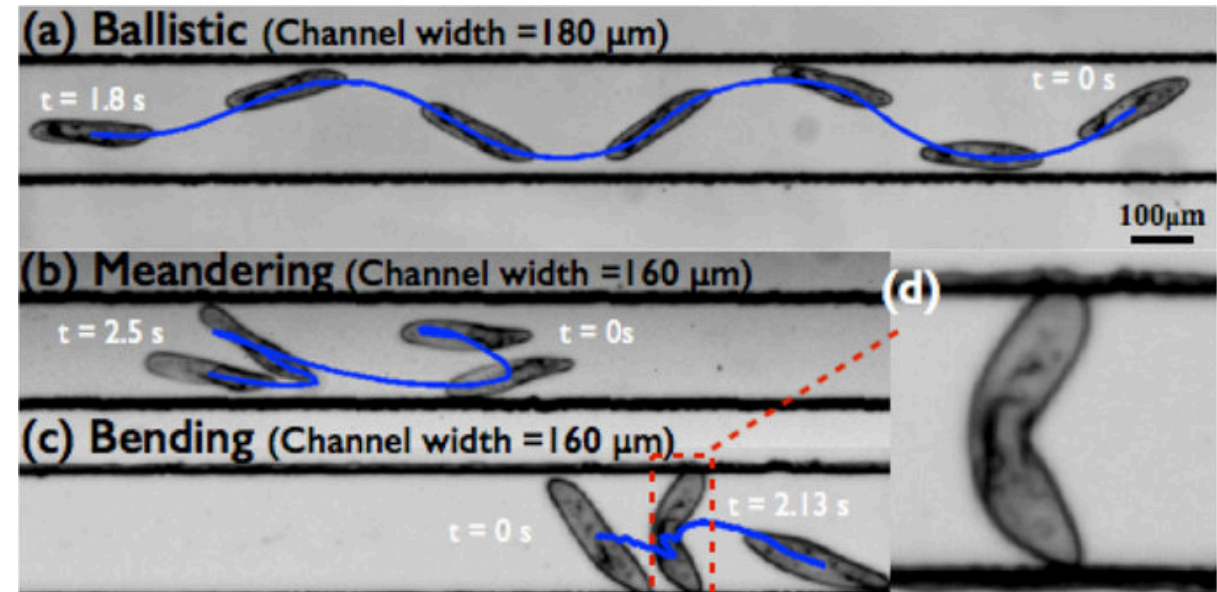


Day 09 - Modeling Drag

Somersault of Paramecium in extremely confined environments -
>

source:

<https://www.nature.com/articles/srep13148>



Announcements

- Homework 3 is posted (due Friday; late after Sunday)
- Homework 4 is posted (due next Friday; late after Sunday)
- **First midterm is coming up** (assigned 16 Feb)
 - One exercise will ask you to get started on your final project planning.
 - *Who are you gonna work with? What are you interested in studying?*
Start thinking about this!
- Limiting using [RaiseMyHand](#) to Workshop Days on (Fridays) only.
 - *Will start posting the link to collect your questions for those days.*

HW1 Feedback

- Overall, folks did well; some issues with uploading/submitting
 - Submit a regrade request if you think something was graded incorrectly
- Kang has **ALOT** of grading to do
- Grading of your HW will be completed by the Monday a week following the Sunday due date
- We will not grade homework that has not been submitted properly
 - Delays will be handled on a case-by-case basis

Please assign problem sections to your assignment. *It takes five more minutes.*

Please submit your assignment as a single PDF file. *It makes sure your*

Clicker Question 9-0

I need help making sure my homework submissions are done correctly.

1. I think I am doing it correctly.
2. I am not sure if I am doing it correctly.
3. I know I am not doing it correctly. Please help.

If you select 2 or 3, make a mental note to please see me soon and so we can go through it together or debug your process.

Weekly Seminars

TUESDAY, February 3, 2026

Nuclear Theory Seminar, 11:00 a.m., FRIB 1200 Lab & Zoom

Speaker: Ryan LaRose, MSU

Title: Error mitigation for partially-error corrected quantum computers

Zoom Link: 964 7281 4717

Passcode: 48824

Weekly Seminars

WEDNESDAY, February 4, 2026

Astronomy Seminar, 1:30 p.m., BPS 1400 & Zoom

Speaker: Marvin Morgan, UCSB

Title: TBA

Zoom Link: <https://msu.zoom.us/j/93334479606?pwd=OtIXPWhRPBfzYu53sl3trSJlaBYI7C.1>

Passcode: 825824

Weekly Seminars

THURSDAY, February 5, 2026

Colloquium, Seminar, 3:30 p.m., BPS 1415 & Zoom

Refreshments at 3:00 BPS in BPS 1400

Speaker: Glennys Farrar, New York University

Title: Origin of ultrahigh-energy cosmic rays in Binary Neutron Star Collisions
and the crucial roles of Nuclear Physics

Zoom Link: <https://msu.zoom.us/j/94951062663>

Password: 2002

Weekly Seminars

FRIDAY, February 6, 2026

QuIC, Seminar, 12:40 p.m., BPS 1300 & Zoom

Speaker: Gregory Quiroz, APL

Title: Classical Non-Markovian Noise in Symmetry-Preserving Quantum Dynamics

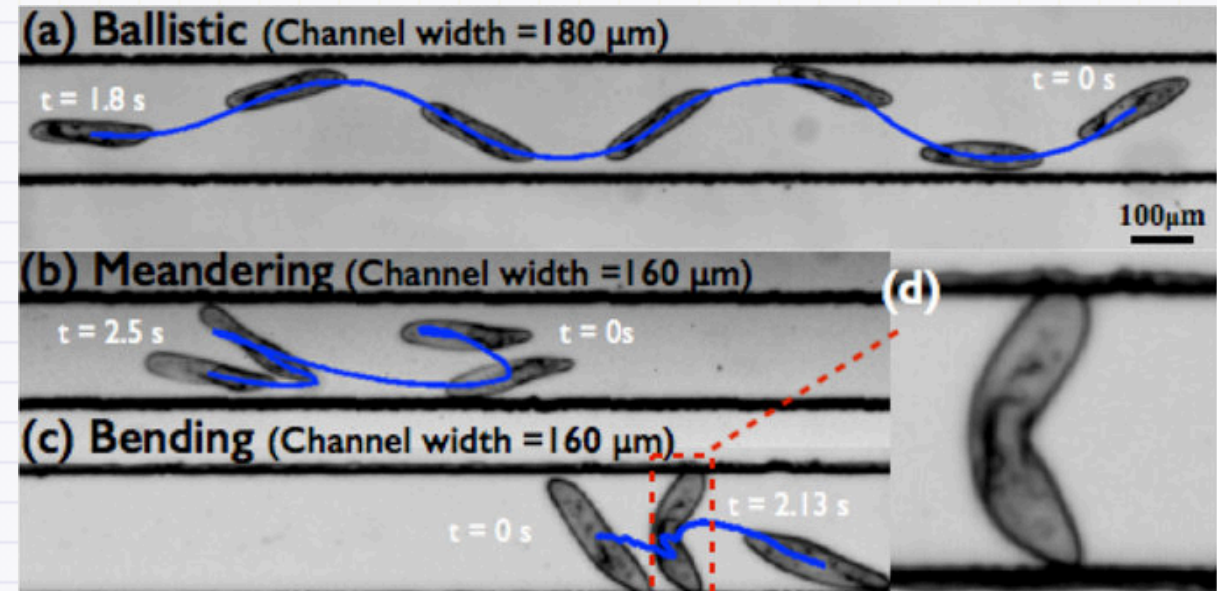
*For the full schedule, please see: <https://sites.google.com/msu.edu/quic-seminar/> or for more information, please reach out to Ryan LaRosa directly

Reminder:

Email Danny (caball14@msu.edu) your extra credit seminar write-ups

Goals for this week

- Establish a model for drag forces
- Develop an understanding of the process for modeling forces
- Produce equations of motion that can be investigated
- Start probing the behavior of these systems with math and computing



Reminders

Force Models

We have been modeling the drag force using a functional dependence on velocity.

$$\vec{F}_{\text{drag}} = -f(v)\hat{v}$$

where $f(v)$ is a function of velocity.

We established (in 1D) there are two common forms of drag force:

$$f(v) = bv \quad \text{Linear Drag}$$

$$f(v) = cv^2 \quad \text{Quadratic Drag}$$

Reminders

Reynolds Number

The choice of which drag model to use depends on the Reynolds number of the system:

$$\text{Re} = \frac{\rho v L}{\eta}$$

where ρ is the density of the fluid, v is the velocity of the object relative to the fluid, L is a characteristic linear dimension (e.g., diameter of a sphere), and η is the dynamic viscosity of the fluid.

Reminders

Equations of Motion

The next step is to use Newton's 2nd Law to write the equations of motion for the system. We found those equation of motion to be:

$$m\dot{v} = -f(v)$$

where $f(v)$ is the drag force. So for each form of drag we have:

$$\dot{v} = g - \frac{b}{m}v \quad \text{Linear Drag}$$

$$\dot{v} = g - \frac{c}{m}v^2 \quad \text{Quadratic Drag}$$

Reminders

Trajectories

We can integrate these equations of motion to find the velocity as a function of time. We found:

$$v(t) = v_{t,\text{lin}} \left(1 - e^{-\frac{bt}{m}} \right) \quad \text{Linear Drag}$$

$$v(t) = v_{t,\text{quad}} \tanh \left(\frac{gt}{v_{t,\text{quad}}} \right) \quad \text{Quadratic Drag}$$

where $v_{t,\text{lin}} = \frac{mg}{b}$ for linear drag and $v_{t,\text{quad}} = \sqrt{\frac{mg}{c}}$ for quadratic drag.

Our Current Investigatory Process

The Model-to-Trajectory Pipeline

1. Model the forces acting on the system
2. Write the equations of motion using Newton's 2nd Law
3. Solve the equations of motion to find trajectories

This is incomplete. We will need to learn how stability, critical points, and phase space can help us understand the behavior of these systems.

We have also only done step 3 analytically. We will need to learn how to use computing to investigate these systems.

Clicker Question 9-1

The Reynolds number can be thought of as a comparison between two forces acting on an object moving through a fluid: (1) inertial forces and (2) viscous forces. The ratio gives a sense of which force dominates the behavior of the object.

$$\text{Re} = \frac{\rho v L}{\eta}$$

Clicker Question 9-1

With this in mind, consider again the Reynolds number formula:

$$\text{Re} = \frac{\rho v L}{\eta},$$

Consider three scenarios:

1. A person swimming through water at moderate speed.
2. A small bacterium swimming through water at low speed.
3. A large airplane flying through the air at high speed.

Order these scenarios from lowest to highest Reynolds number.

1. 1, 3, 2; **2.** 2, 3, 1; **3.** 1, 2, 3, **4.** 3, 1, 2; **5.** Something else

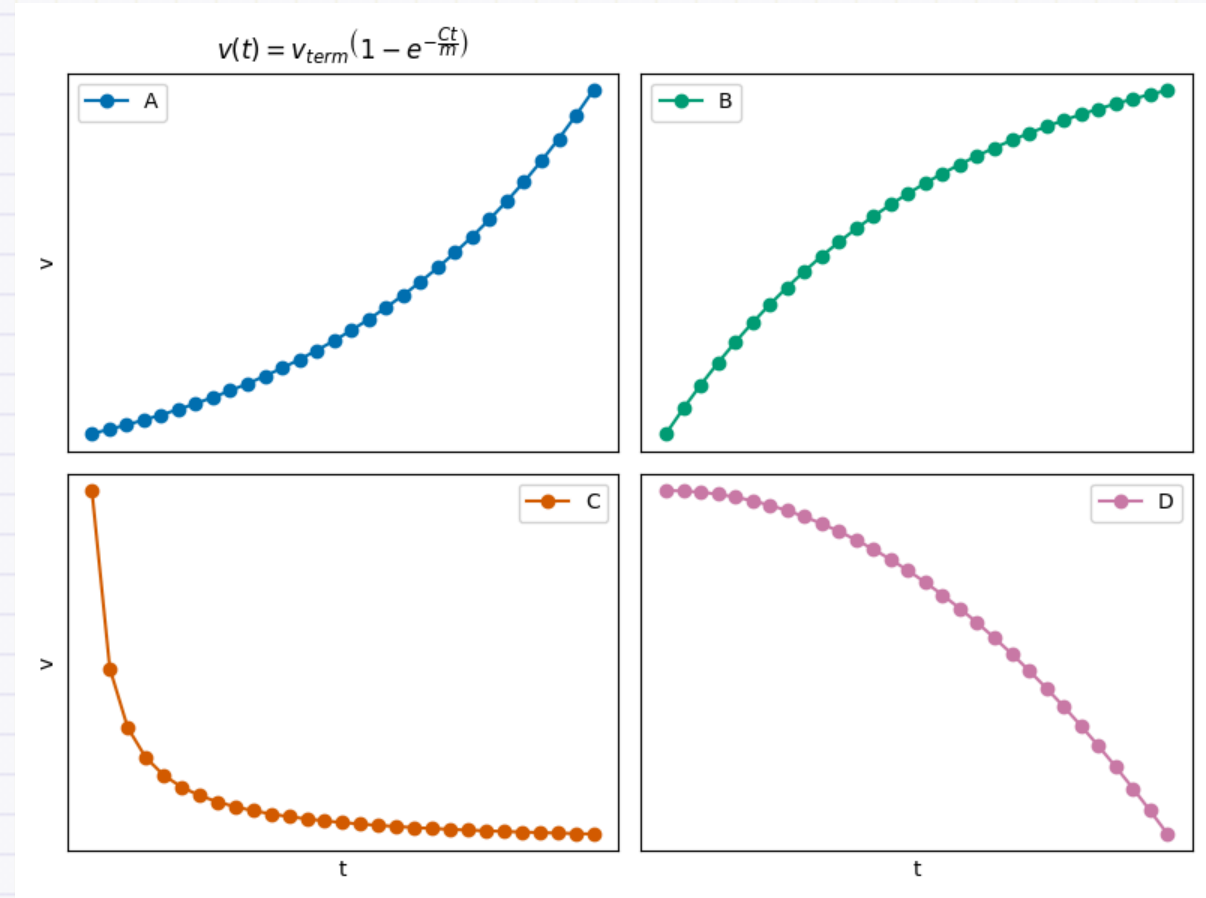
Clicker Question 9-2

For the system of **Linear Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \left(1 - e^{-\frac{bt}{m}} \right)$$

where $v_{term} = \sqrt{\frac{mg}{b}}$.

Which sketch could be correct for the velocity of the ball?



Clicker Question 9-3

For the system of **Quadratic Drag in 1D**, we found a solution for the velocity as a function of time, with $v = 0$ at $t = 0$.

$$v(t) = v_{term} \tanh(gt/v_{term})$$

where $v_{term} = \sqrt{mg/c}$. What happens when $t \rightarrow \infty$?

1. The object stops moving.
2. The object travels at a constant velocity.
3. The object travels at an increasing velocity.
4. The object travels at a decreasing velocity.
5. I'm not sure.

Clicker Question 9-4

For quadratic drag in 2D, we found the following pair of differential equations:

$$\dot{v}_x = -\tilde{D}v_x\sqrt{v_x^2 + v_y^2}$$
$$\dot{v}_y = -\tilde{D}v_y\sqrt{v_x^2 + v_y^2} - g$$

True or False: This pair of differential equations can be decoupled.

1. True
2. False
3. ???

Clicker Question 9-5

For linear drag in 2D, we found the following pair of differential equations:

$$\begin{aligned}\dot{v}_x &= -\gamma v_x \\ \dot{v}_y &= -\gamma v_y - g\end{aligned}$$

True or False: This pair of differential equations is decoupled.

1. True
2. False
3. ???

Clicker Question 9-6

For the gravitational interaction, I want to compute the force acting on body B, located at $\vec{r}_B$, by body A, located at $\vec{r}_A$.

The gravitational force is given by:

$$\vec{F} = -G \frac{m_1 m_2}{r^2} \hat{r}$$

What is the appropriate form of $\vec{r}$?

1. $\vec{r} = \vec{r}_A - \vec{r}_B$
2. $\vec{r} = \vec{r}_B - \vec{r}_A$
3. Either is ok

Clicker Question 9-7

We found that the equation of motion for the spring-mass system was:

$$\ddot{x} = -\frac{k}{m}x = -\omega^2 x$$

Your friends have proposed the following **general solutions**:

1. $x(t) = A \cos(\omega t)$ 2. $x(t) = B \sin(\omega t)$ 3. $x(t) = A \cos(\omega t) + B \sin(\omega t)$
4. $x(t) = A \cos(\omega t + \phi)$ 5. $x(t) = B \sin(\omega t + \phi)$ 6. $x(t) = A \cos(\omega t + \phi) + B \sin(\omega t + \phi)$

How many of them are correct?

- (1) Only one (2) Two (3) Three
(4) Four (5) All of them